

Joan Escamilla

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Software Engineer II specializing in AI agent and MCP infrastructure engineering, 3+ years building production systems, now building the AI agents and MCP tooling that run them

Experience

RDE Systems | Software Engineer II

January 2023 - Present

- Subject-matter expert on a core module of an HIV/AIDS electronic medical records platform, owning data-entry screens, reports, and migrations across four framework transitions.
- Engineered an autonomous WCAG-compliance agent that scans a codebase and generates per-file SVN patches for human review; validated patch generation on live folders, with full-system deployment projected to cut a 50-to-60-hour manual audit to about 5 hours across 50+ systems.
- Architected the company's first MCP server infrastructure, giving agents direct tool access to our SQL database and SVN history; led a cross-functional meeting with AWS engineers to scope hosting via AgentCore, Bedrock, and Lambda, resolving the missing LLM API endpoint within existing budget.
- Shipped a second agent that automates the team's unwritten SQL conventions, cutting routine script time in half and complex schema-design work by a similar margin.
- Directed an AI-guided rebuild of a 30-year-old product UI and, after a leadership demo, the company's internal site: a reusable, tokenized CSS system shipped as a new frontend in 3 days; now leading a formal design system with 2 junior developers and a 2-week delivery timeline.
- Selected and mentored a 5-developer university capstone team, hiring one into a junior role and continuing to mentor him a year into the job.
- Overhauled the RDE onboarding program, cutting ramp-up time for four new developers joining the framework.

The Coder School | Programming Tutor

September 2021 - July 2023

- Designed and taught a project-based curriculum in React JS, Python, and Java, giving students hands-on experience completing multiple real projects per month.
- Tracked progress across 20+ students with regular parent feedback; students averaged a 30% improvement in programming skills by internal metrics.

MetroStar | Frontend Developer Intern

June 2022 - August 2022

- Designed and built interactive Tableau dashboards using U.S. Census data, giving city planners and officials in Bloomington, Indiana a faster way to identify economic and urban development opportunities.
- Converted Tableau's JS API into reusable React components, cutting dashboard integration time for new use cases across the team.

Education

New Jersey Institute of Technology | B.S Computer Science

May 2023

Certifications

AI Fluency: Framework & Foundations | Anthropic

April 2026

Teaching AI Fluency | Anthropic

April 2026

Introduction to Model Context Protocol | Anthropic

July 2026

Skills

Programming Languages: ColdFusion, SQL, PHP, Python, JavaScript, CSS, HTML

AI/Agentic Development: Agentic AI, MCP Server Development, Workflow Automation, Prompt Engineering, Anthropic Claude, GitHub Copilot

Tools/Frameworks: React JS, jQuery, SQL Server, SVN, Tableau, Figma, Tailwind CSS, Bootstrap, Git, AWS (AgentCore, Bedrock, Lambda)